

TECHNICAL BULLETIN**DIXIE CHEMICAL**

10601 Bay Area Blvd.

Pasadena, TX 77507

Tel: (281) 474-3271

E-Mail: info@dixiechemicalgroup.com**EPOXY CURING AGENT 1000L (Methyltetraphthalic Anhydride)
(ECA 1000L, MTHPA, CAS# 34090-76-1)****Typical Properties**

Properties	Value	Test Method
Molecular Weight	166	NA
Appearance	Clear light to yellow liquid	D-104 (Visual)
Pour Point, °C	-27	ASTM D-97
MTHPA, %	99 min.	D-160
MAN, %	0.1 max.	D-160
Volatiles, ppm	100 max.	D-160A
Free Acid, %	2.0 max.	D-160B-2

Applications

ECA 1000L is a clear light to yellow liquid with a pour point at about -27°C. It is used in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. ECA 1000L performs well when used in demanding applications, such as filament winding, pultrusion, and other forms of lamination, as well as electrical encapsulating, insulating, casting, and potting.

For more information on the use of anhydrides like ECA 1000L as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems", available from Dixie Chemical.

TECHNICAL BULLETIN ECA 1000L (MTHPA)

Formulation Example

ECA 1000L was formulated with a variety of epoxy resins. The samples were formulated with 100 parts of epoxy resin, 83-116 parts ECA 1000L, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 1000L, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	84	590	32	139
Low Viscosity BPA Liquid	87	500	34	141
Cycloaliphatic Epoxy	116	90	41	220
Epoxy Phenol Novolac	92	770	33	137
Epoxy BPA Novolac	83	1630	27	148

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

ECA 1000L will react with water to form diacids. This is normally undesirable, so ECA 1000L should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 1000L, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

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ECA 1000L (MTHPA)**Health Hazards**

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes.

Health Hazards (cont.)

In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 1000L is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.